

History of Deep Learning

Dr. Zulfikar Ali Ansari

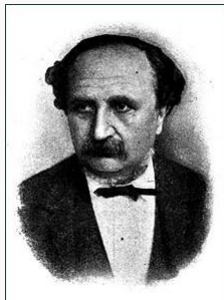
Acknowledgements

- Most of this material is based on the article "Deep Learning in Neural Networks: An Overview" by J.Schmidhuber^[1]

Biological Neurons

Reticular Theory

Joseph von Gerlach proposed that the nervous system is a single continuous network as opposed to a network of many discrete cells!



1871-1873

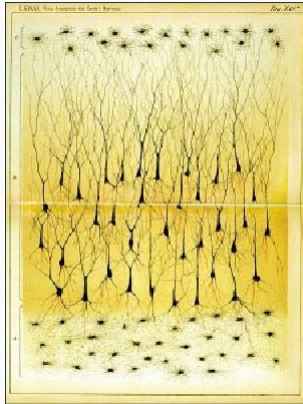


Reticular theory

Staining Technique

Camillo Golgi discovered a chemical reaction that allowed him to examine nervous tissue in much greater detail than ever before

He was a proponent of Reticular theory.



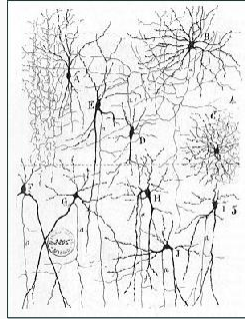
1871-1873



Reticular theory

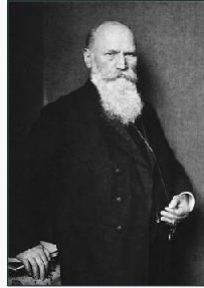
Neuron Doctrine

Santiago Ramón y Cajal used Golgi's technique to study the nervous system and proposed that it is actually made up of discrete individual cells forming a network (as opposed to a single continuous network)



The Term Neuron

The term neuron was coined by Heinrich Wilhelm Gottfried von Waldeyer-Hartz around 1891.

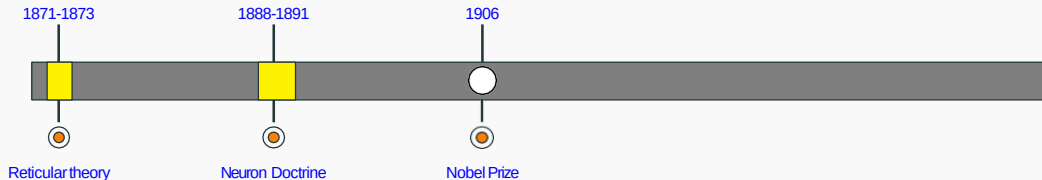
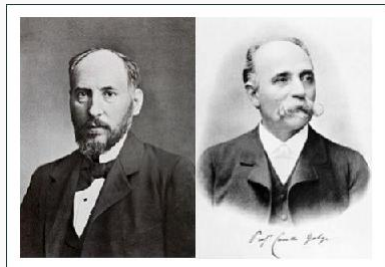


He further consolidated the Neuron Doctrine.



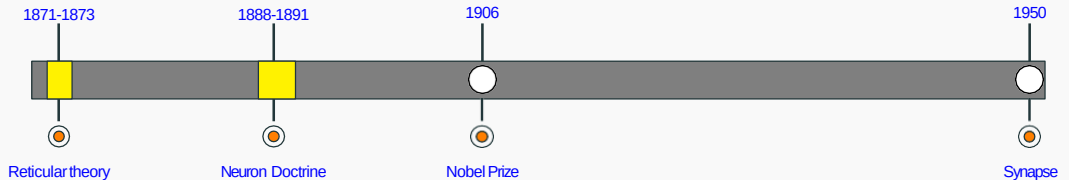
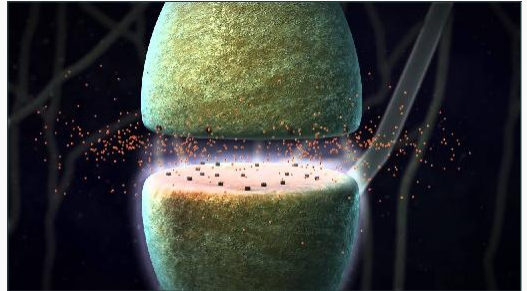
Nobel Prize

Both Golgi (reticular theory) and Cajal (neuron doctrine) were jointly awarded the 1906 Nobel Prize for Physiology or Medicine, that resulted in lasting conflicting ideas and controversies between the two scientists.



The Final Word

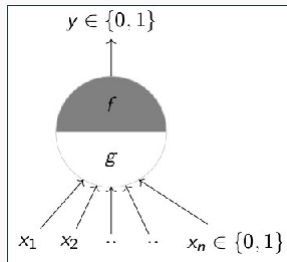
In 1950s electron microscopy finally confirmed the neuron doctrine by unambiguously demonstrating that nerve cells were individual cells interconnected through synapses (a network of many individual neurons).



From Spring to Winter of AI

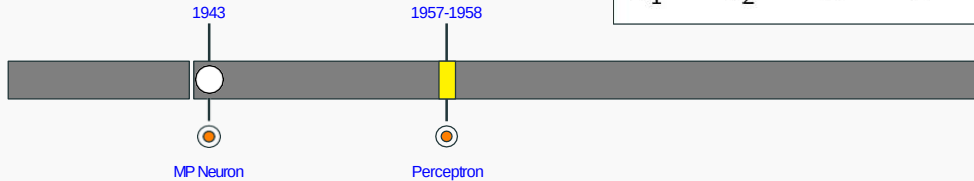
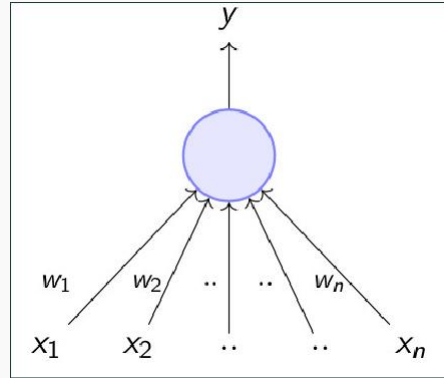
McCulloch Pitts Neuron

McCulloch (neuroscientist) and Pitts (logician) proposed a highly simplified model of the neuron (1943) [\[2\]](#)



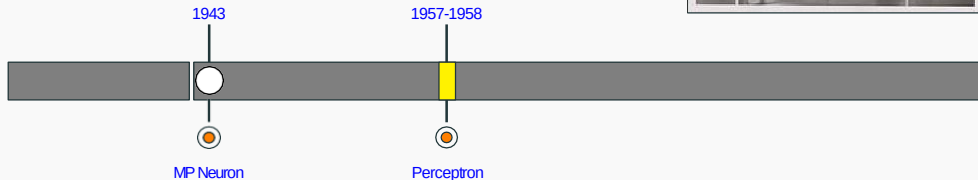
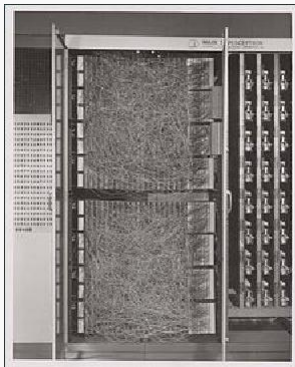
Perceptron

“the perceptron may eventually be able to learn, make decisions, and translate languages” -Frank Rosenblatt



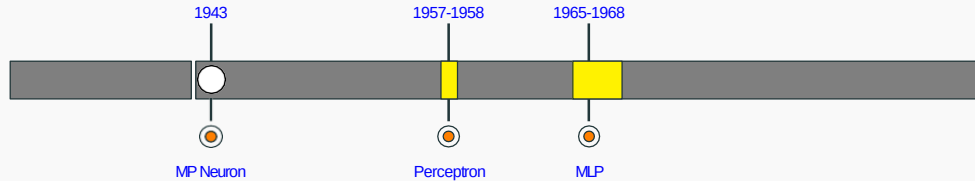
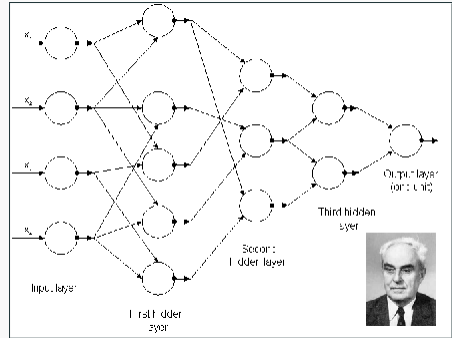
Perceptron

“the embryo of an electronic computer that the Navy expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.” -New York Times



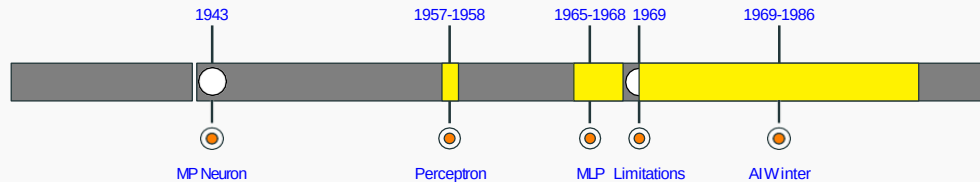
First generation Multilayer Perceptrons

Ivakhnenko et. al. [\[3\]](#)



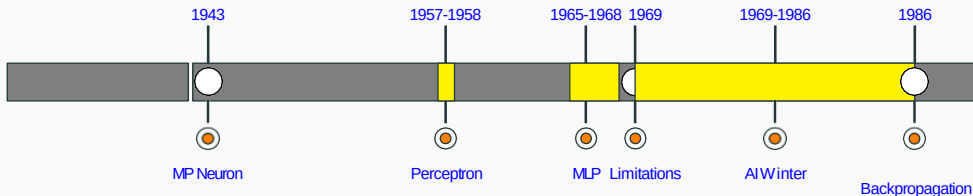
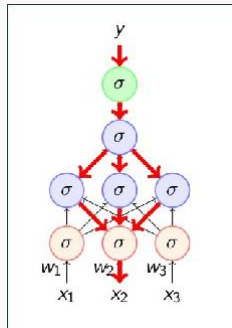
AI Winter of connectionism

Almost lead to the abandonment of
connectionist AI



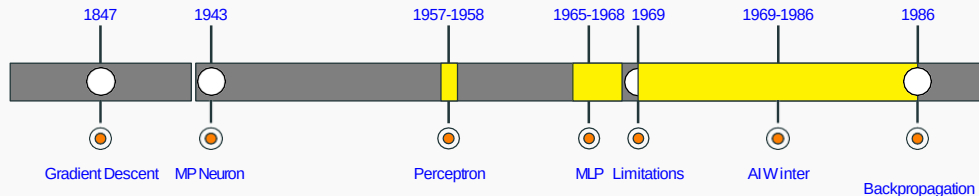
Backpropagation

- Discovered and rediscovered several times throughout 1960's and 1970's
- Werbos(1982)^[5] first used it in the context of artificial neural networks
- Eventually popularized by the work of Rumelhart et. al. in 1986^[6]



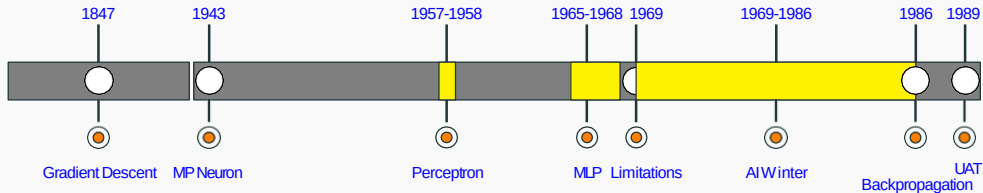
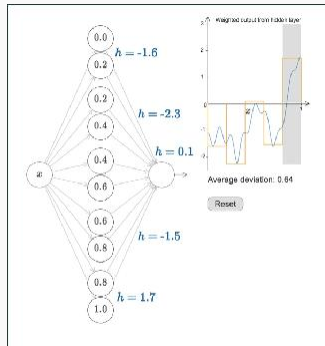
Gradient Descent

Cauchy discovered Gradient Descent
motivated by the need to compute the orbit
of heavenly bodies



Universal Approximation Theorem

A multilayered network of neurons with a single hidden layer can be used to approximate any continuous function to any desired precision^[7]

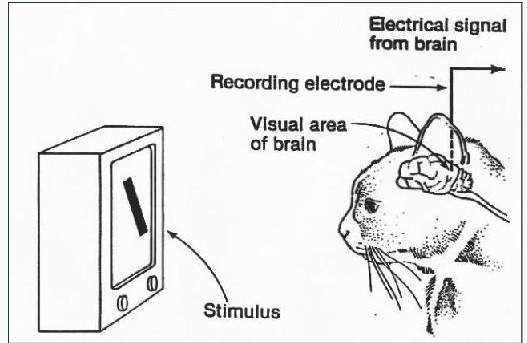


The Deep Revival

From Cats to Convolutional Neural Networks

Hubel and Wiesel Experiment

Experimentally showed that each neuron has a fixed receptive field - i.e. a neuron will fire only in response to a visual stimuli in a specific region in the visual space [\[18\]](#)



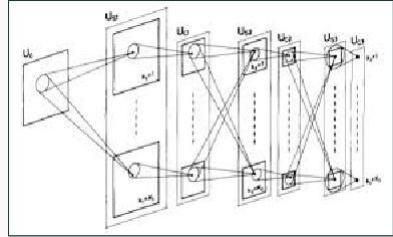
1959



Hand W experiment

Neocognitron

Used for Handwritten character recognition and pattern recognition (Fukushima et. al.)[\[19\]](#)



1959



Hand and W experiment

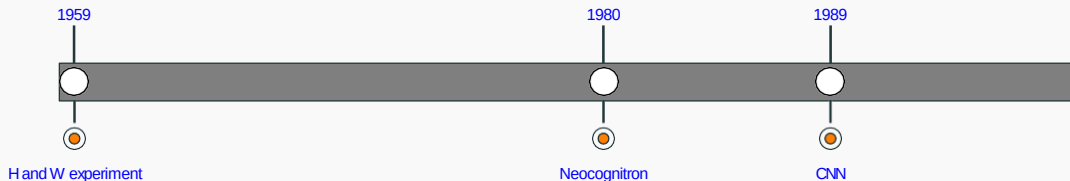
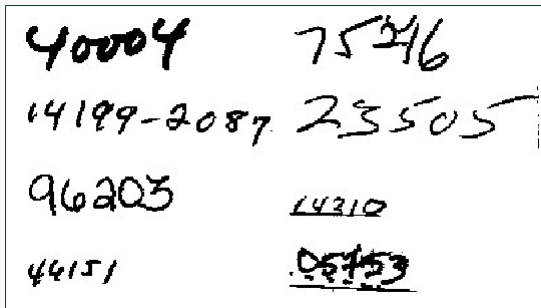
1980



Neocognitron

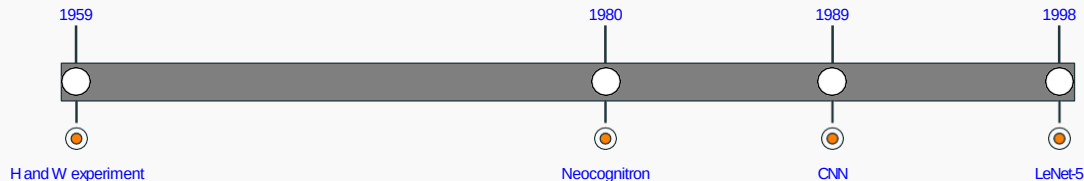
Convolutional Neural Network

Handwriting digit recognition using
backpropagation over a Convolutional Neural
Network (LeCun et. al.) [\[20\]](#)



LeNet-5

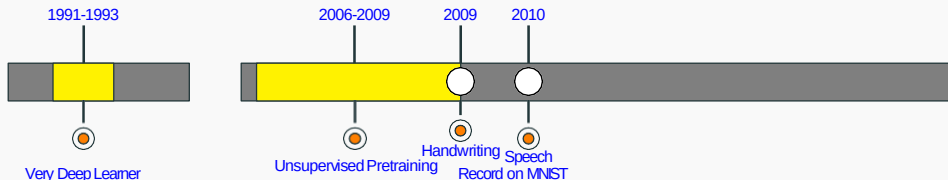
Introduced the (now famous) MNIST dataset
(LeCun et. al.) [\[21\]](#)



New record on MNIST

Ciresan et. al. set a new record on the MNIST dataset using good old backpropagation on GPUs (GPUs enter the scene) [\[11\]](#)

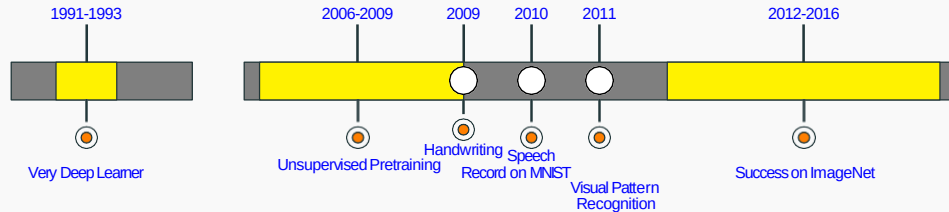
2 ² 17	1 ¹ 71	9 ⁸ 98	9 ⁹ 59	9 ⁹ 79	5 ⁵ 35	3 ⁸ 23
4 ⁹ 49	5 ⁵ 35	9 ⁴ 97	4 ⁹ 49	4 ⁴ 94	0 ² 02	5 ⁵ 35
6 ⁶ 16	4 ⁴ 94	0 ⁶ 60	6 ⁶ 06	6 ⁶ 86	1 ¹ 79	1 ¹ 71
9 ⁹ 49	0 ⁰ 50	5 ⁵ 35	8 ⁸ 98	7 ⁹ 79	7 ⁷ 17	1 ¹ 61
2 ⁷ 27	8 ⁸ 58	7 ² 78	6 ⁶ 16	6 ⁵ 65	4 ⁴ 94	0 ⁰ 60



Winning more visual recognition challenges



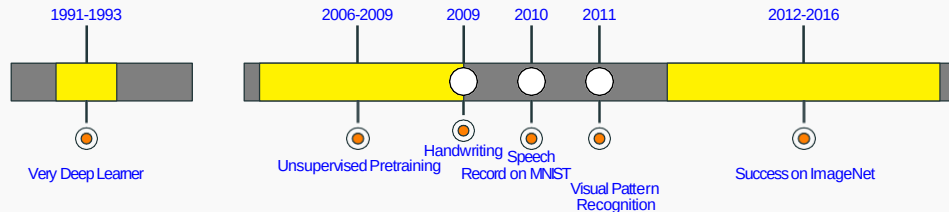
Network	Error	Layers
AlexNet ^[13]	16.0%	8



Winning more visual recognition challenges



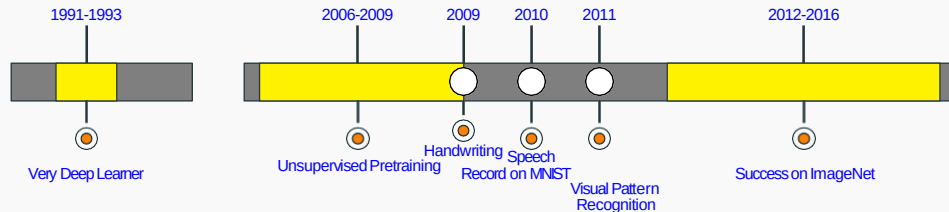
Network	Error	Layers
AlexNet ^[13]	16.0%	8
ZFNet ^[14]	11.2%	8



Winning more visual recognition challenges



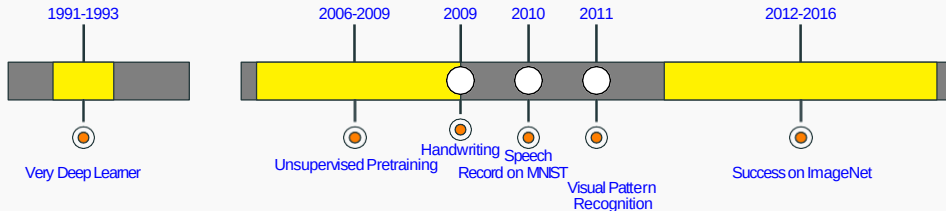
Network	Error	Layers
AlexNet ^[13]	16.0%	8
ZFNet ^[14]	11.2%	8
VGGNet ^[15]	7.3%	19



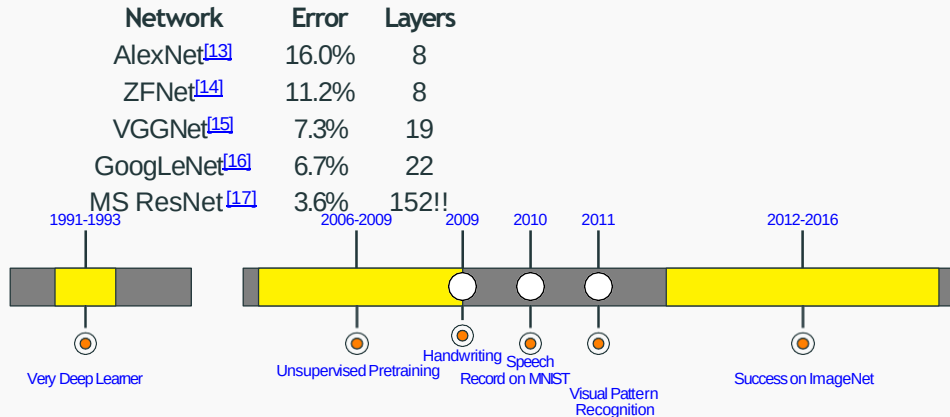
Winning more visual recognition challenges



Network	Error	Layers
AlexNet ^[13]	16.0%	8
ZFNet ^[14]	11.2%	8
VGGNet ^[15]	7.3%	19
GoogLeNet ^[16]	6.7%	22



Winning more visual recognition challenges

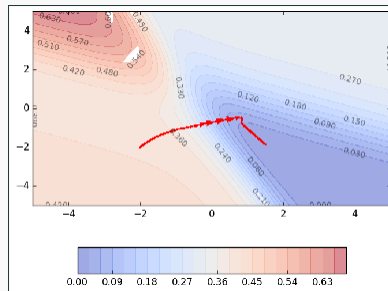


An algorithm inspired by an experiment on cats is today used to detect cats in videos :-)

Faster, higher, stronger

Better Optimization Methods

Faster convergence, better accuracies



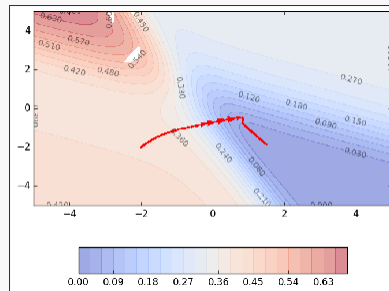
1983



Nesterov

Better Optimization Methods

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1983



Nesterov

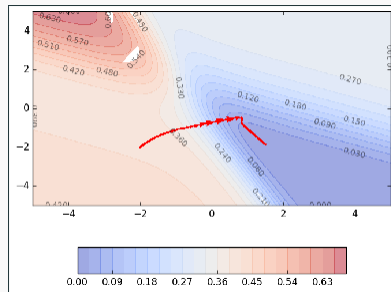
2011



Adagrad

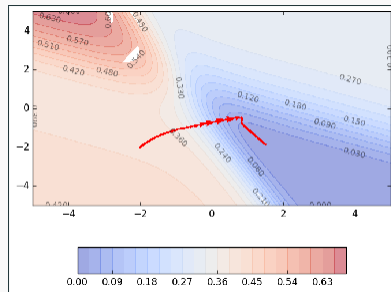
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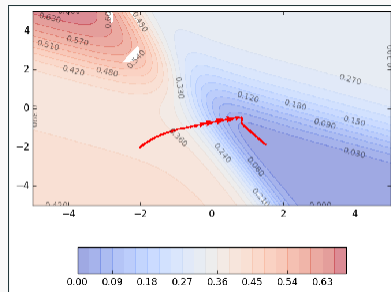
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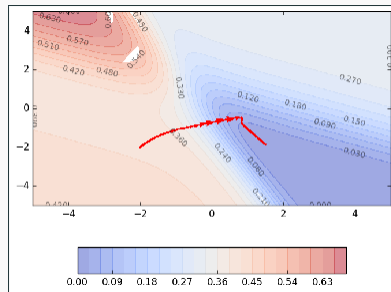
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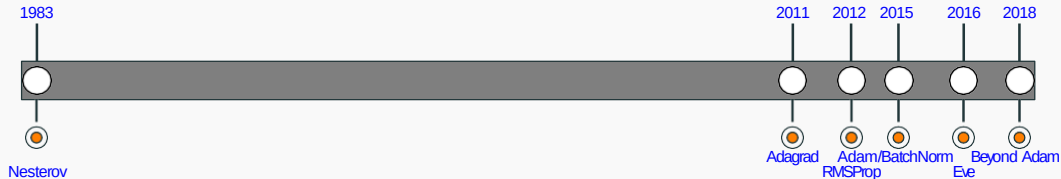
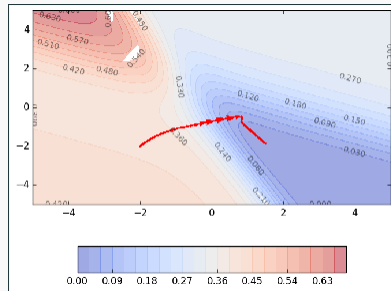
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Better Optimization Methods

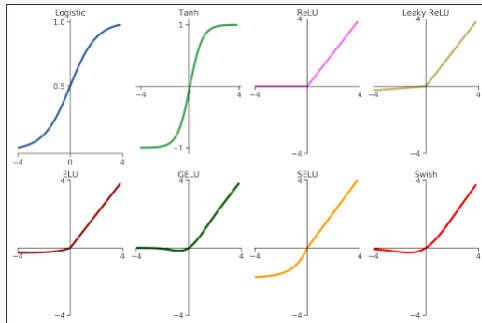
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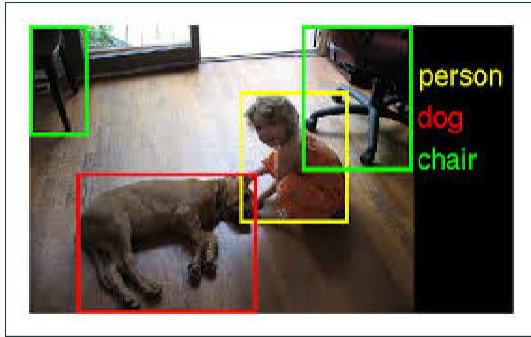
Better Activation Functions

We have come a long way from the initial days when the logistic function was the default activation function in NNs!

Over the past few years many new functions have been proposed leading to better convergence and/or performance!



Object Detection/Recognition



- Semantic Segmentation (Long et al, 2015) [\[59\]](#)
- Recurrent CNNs (Liang et al., 2015) [\[60\]](#)
- Faster RCNN (Ren et al., 2015) [\[61\]](#)
- Inside-Outside Net (Bell et al., 2015) [\[62\]](#)
- YOLO9000 (Redmon et al., 2016) [\[63\]](#)
- R-FCN (Dai et al., 2016) [\[64\]](#)
- Mask R-CNN (He et al., 2017) [\[65\]](#)
- Video Object segmentation (Caelles et al., 2017) [\[66\]](#)



Visual Tracking

- Choi et al. (2017) [\[67\]](#)
- Yun et al. (2017) [\[68\]](#)
- Alahi et al. (2017) [\[69\]](#)

Retr.



1. Top view of the lights of a city at night, with a well-illuminated square in front of a church in the foreground;
2. People on the stairs in front of an illuminated cathedral with two towers at night;



1. Tourists are sitting at a long table with beer bottles on it in a rather dark restaurant and are raising their bierglaeser;
2. Tourists are sitting at a long table with a white table-cloth in a somewhat dark restaurant;

Gen.

A square with burning street lamps and a street in the foreground;

Tourists are sitting at a long table with a white table cloth and are eating;

Image Captioning

- Mao et al. (2014) [\[70\]](#)
- Mao at al. (2015) [\[71\]](#)
- Kiros et al. (2015) [\[72\]](#)
- Donahue et al. (2015) [\[73\]](#)
- Vinyals et al. (2015) [\[74\]](#)
- Karpathy et al. (2015) [\[75\]](#)
- Fang et al. (2015) [\[76\]](#)
- Chen et al. (2015) [\[77\]](#)



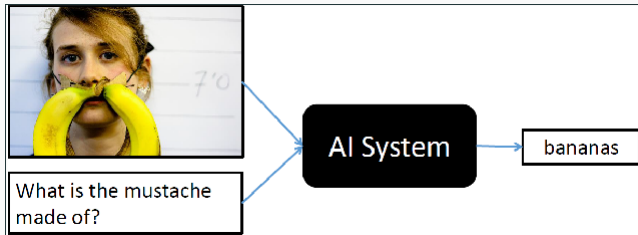
A group of young men playing a game of soccer



A man riding a wave on top of a surfboard.

Video Captioning

- Donahue et al. (2014) [\[78\]](#)
- Venugopalan et al. (2014) [\[79\]](#)
- Pan et al. (2015) [\[80\]](#)
- Yao et al. (2015) [\[81\]](#)
- Rohrbach et al. (2015) [\[82\]](#)
- Zhu et al. (2015) [\[83\]](#)
- Cho et al. (2015) [\[34\]](#)



Visual Question Answering

- Santoro et al. (2017) [\[84\]](#)
- Hu et al. (2017) [\[85\]](#)
- Johnson et al. (2017) [\[86\]](#)
- Ben-younes et al. (2017) [\[87\]](#)
- Malinowski et al. (2017) [\[88\]](#)
- Kazemi et al. (2016) [\[89\]](#)

She _____.



(nods)

She opens the _____.



(door)



Question: What is the cat doing? Answer: playing with a tablet

Video Question Answering

- Tapaswi et. al. 2016 [\[90\]](#)
- Zeng et. al. 2016 [\[91\]](#)
- Maharaj et. al. 2017 [\[92\]](#)
- Zhao et. al. 2017 [\[93\]](#)
- Yu Youngjae et. al. 2017 [\[94\]](#)
- Xue Hongyang et. al. 2017 [\[95\]](#)
- Mazaheri et. al. 2017 [\[96\]](#)

Input video



Summary



Video Summarization

- Chheng 2007 [\[97\]](#)
- Ajmal 2012 [\[98\]](#)
- Zhang Ke 2016 [\[99\]](#)
- Zhong J 2017 [\[100\]](#)
- Panda 2017 [\[101\]](#)



Generating Authentic Photos

Variational Autoencoders

(Kingma et. al., 2013) [\[102\]](#)

Generative Adversarial

Networks (Goodfellow et. al.,
2014) [\[103\]](#)

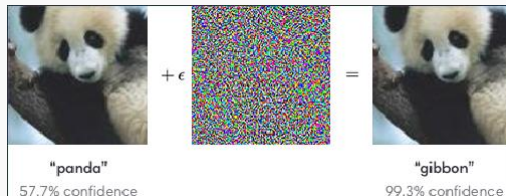
Plug & Play generative nets
(Nguyen et al., 2016) [\[104\]](#)

Progressive Growing of GANs
(Karras et al., 2017) [\[105\]](#)

Calls for Sanity (Interpretable, Fair, Responsible, Green AI)

The Paradox of Deep Learning

Why does deep learning work so well despite

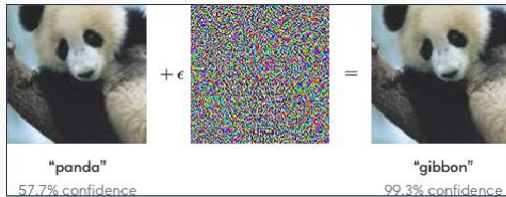


*<https://arxiv.org/pdf/1710.05468.pdf>

The Paradox of Deep Learning

Why does deep learning work so well despite

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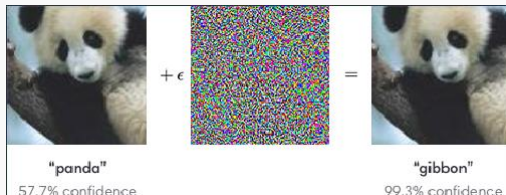


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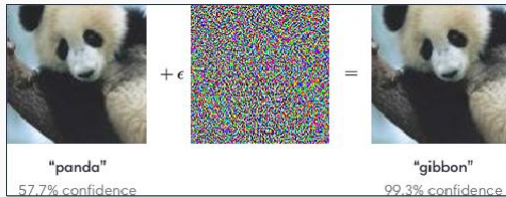


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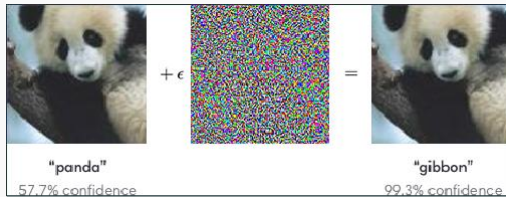


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- non-robustness (see figure)



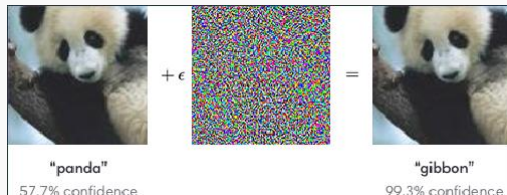
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No clear answers yet but ...



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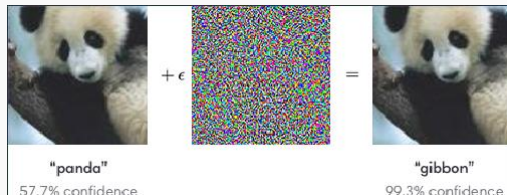
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No clear answers yet but ...

- Slowly but steadily there is increasing emphasis on explainability and theoretical justifications!*

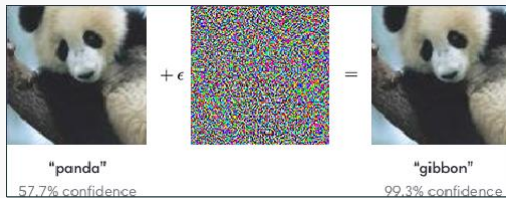


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No clear answers yet but ...

- Slowly but steadily there is increasing emphasis on explainability and theoretical justifications!*
- Hopefully this will bring sanity to the proceedings !

*<https://arxiv.org/pdf/1710.05468.pdf>

Tell me why!

Workshop on Human Interpretability in
Machine Learning

*We still do not know much about why DL models
do what they do!*



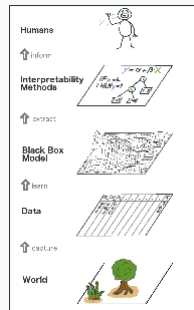
2016



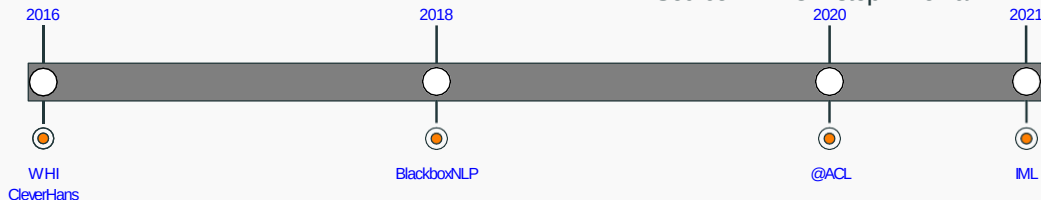
WHI

Tell me why!

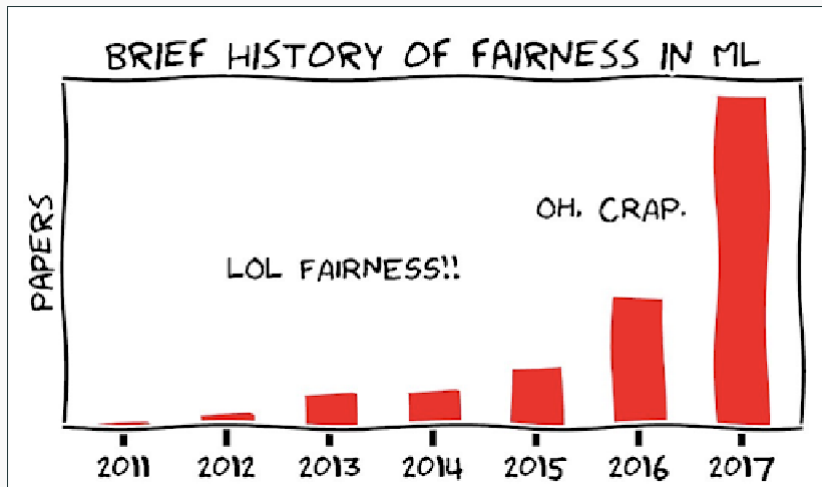
*Interpretable Machine Learning: A Guide for
Making Black Box Models Explainable. —
Christoph Molnar*



Source: IML: Christoph Molnar



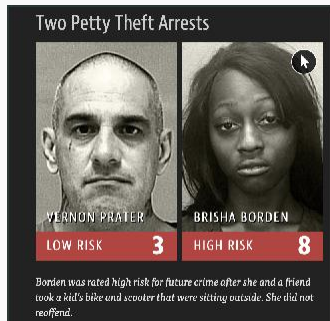
Be Fair and Responsible!



Source: <https://fairmlclass.github.io/> (Moritz Hardt)

Be Fair and Responsible!

“There’s software used across the country to predict future criminals. And it’s biased against blacks.” - Propublica



Source:

<https://www.propublica.org/article/machine-bias->

2016



Machine Bias

Be Fair and Responsible!

“Facial Recognition Is Accurate, if You’re a White Guy” - MIT Media

Gender Shades audit, 2018

Accuracy in gender classification

	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
IBM	88.0%	65.3%	99.7%	92.5%	34.4%
Megvii	95.3%	65.5%	99.2%	94.0%	33.8%
Microsoft	94.0%	79.2%	100.0%	98.3%	20.8%

Chart: MIT Technology Review • Source: Joy Buolamwini & Timnit Gebru • Created with Datawrapper

Oprah Winfrey



amazon

Source: Joy Buolamwini (Youtube)

2016



Machine Bias

2018

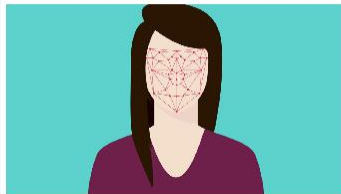


Gender/Race Bias

Be Fair and Responsible!

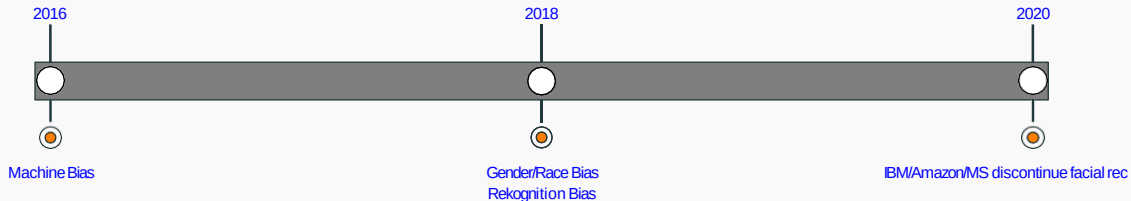
Microsoft refuses to sell police its facial-recognition technology, following similar moves by Amazon and IBM

IBM says it is no longer working on face recognition because it's used for racial profiling



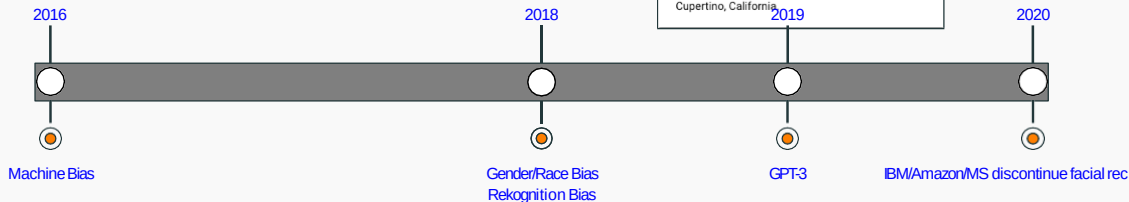
Technology

Microsoft won't sell police its facial-recognition technology, following similar moves by Amazon and IBM



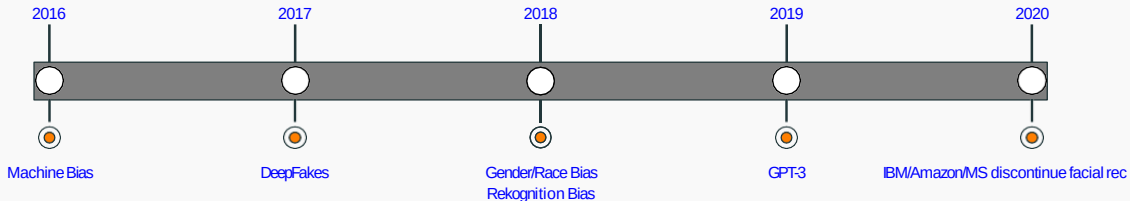
Be Fair and Responsible!

“Due to our concerns about malicious applications of the technology, we are not releasing the trained model.” — *OpenAI*



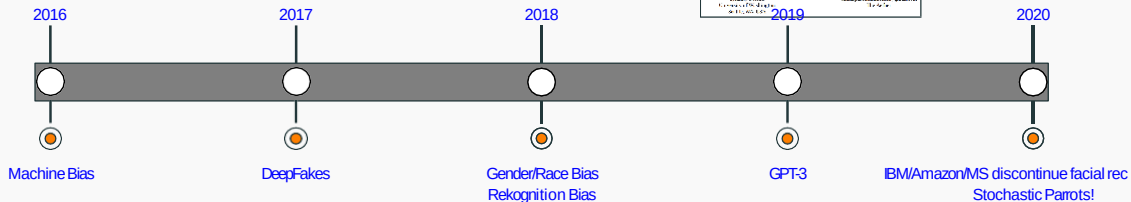
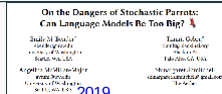
Be Fair and Responsible!

What started off as an innocuous project for mimicking facial expressions has since lead to many apps and creation of fake videos for blackmailing, pronography and swaying elections!



Be Fair and Responsible!

“Models are only as good as the data. Be responsible while curating data.” – *Bender et al.*



Push for Green AI

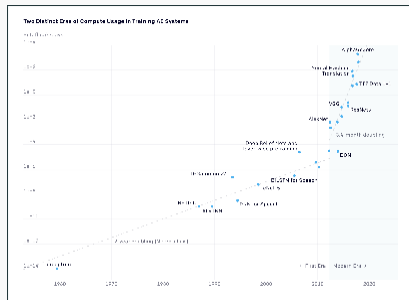
The computations required for deep learning research have been doubling every few months, resulting in an estimated 300,000x increase from 2012 to 2018 – *AllenAI*

Ironically, deep learning was inspired by the human brain, which is remarkably energy efficient.

2019



GreenAI



<https://openai.com/blog/ai-and-compute/>

Push for Green AI

Call for energy and policy considerations for Deep Learning

	Date of original paper	Energy consumption (kWh)	Carbon footprint (lbs of CO2e)	Cloud compute cost (USD)
Transformer (Vaswani et al.)	Mar 2017	78	79	\$11.1141
Transformer (270M parameters)	Mar 2017	263	140	\$144.0413
EvoE	Feb 2018	276	232	\$459.8472
MIT: 175M parameters	Oct 2018	507	485	\$972.112.971
Trajectory (270M parameters) w/ 100M parameters	Mar 2017	390.347	620.115	\$946.79.112.112
GPT2	Feb 2019	-	-	\$1.2025-043.010

Note: We converted kWh to lbs of CO2e using the conversion factor 1 kWh = 0.000476 lbs of CO2e. The cloud compute cost is based on the current prices of AWS EC2 instances. The data is sourced from the paper: <https://arxiv.org/abs/2004.05909>

Common carbon footprint benchmarks

in lbs of CO2 equivalent

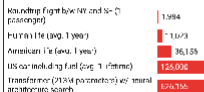


Chart: MIT Technology Review - Source: Ghemawat et al. - Created with Datawrapper

2019



GreenAI

Energy-Aware NLP

<https://github.com/ChristosChristofidis/awesome-deep-learning>

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